



ASSIGNMENT NO.1

OBJECT ORIENTED PROGRAMMING

Submitted by-

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2034

Q1. Write a program (WAP) to display "Hello World" on console display

```
#include <iostream> using
namespace std; int main()
{ cout<<"Hello World";

    return 0;
}
```

```
Output
Hello World
=== Code Execution Successful ===
```

Q2. WAP to input an integer, decimal and character and display it.

```
#include <iostream> using
namespace std; int main()
{ int a ;
  cout<<"Enter a number : ";
  cin>>a; cout<<a;

  return 0;
}
```

```
Output
Enter a number : 7
7
=== Code Execution Successful ===
```

```
#include <iostream> using
namespace std; int main() {
float a ;
  cout<<"Enter a number : ";
  cin>>a; cout<<a;

  return 0; }
```

```
Output
Enter a number : 8.6
8.6
=== Code Execution Successful ===
```

```
#include <iostream> using namespace std; int main() { char str[20]; cout<<"Enter a
character : "; cin>>str; cout<<str;
```

```
return 0; }
```

Q3. WAP to add, subtract, multiply, divide numbers.

```
#include <iostream> using namespace std; int main()
{ int a; int b; int sum; cout<<"Enter first
number:"<<endl; cin>>a; cout<<"Enter second
number : "<<endl; cin>>b; sum=a+b; cout<<"sum
is "<<sum<<endl;
return 0;
}
```

```
#include <iostream> using namespace std; int main()
{ int a; int b; int subtract; cout<<"Enter first
number:"<<endl; cin>>a; cout<<"Enter second
number : "<<endl; cin>>b; subtract=a-b; cout<<
"subtract is "<<subtract<<endl; return 0;
}
```

Output

```
Enter a character : Saanvi
Saanvi

=== Code Execution Successful ===
```

Output

```
Enter first number:
8
Enter second number :
9
sum is 17
|

=== Code Execution Successful ===
```

Output

```
Enter first number:
8
Enter second number :
7
subtract is 1

=== Code Execution Successful ===
```

```
#include <iostream> using namespace std; int main() { int a; int b; int product; cout<<"Enter
first number:"<<endl; cin>>a; cout<<"Enter second number : "<<endl; cin>>b; product=a*b;
cout<< "product is "<<product<<endl;
    return 0;
}
```

Output

```
Enter first number:
5
Enter second number :
6
product is 30
```

=== Code Execution Successful ===

```
#include <iostream>
using namespace std; int
main()
{ int a;
int b; int
second
cout<<
    return
}
```

Output

```
Enter first number:
8
Enter second number :
4
division is 2
```

=== Code Execution Successful ===

```
division; cout<<"Enter first
number:"<<endl; cin>>a; cout<<"Enter
number : "<<endl; cin>>b; division=a/b;
"division is "<<division<<endl;
0;
```

Q4. Write a C++ program that will ask for a temperature in Celsius and display it in degree Fahrenheit.[$F = 9C/5 + 32$]

```
#include <iostream> using namespace std; int main()
{ float C; float F; cout<<"Enter temp in Celsius ";
cin>>C;
F=((9*C)/5)+32; cout<<"Temp in
Fahrenheit is "<<F<<endl;
```

Output

```
Enter temp in Celsius 98
Temp in Fahrenheit is 208.4
```

=== Code Execution Successful ===

```

    return 0;
}

```

Q5. WAP to format console output using '\n', '\t', '\b' , endl, setw within cout statement

```

#include <iostream> using namespace std; int main()
{   char str[20];   int g;
    cout<<"Enter your name:\n";   cin>>str;
    cout<<"Enter your group:\n ";   cin>>g;
    cout<<str<<"\n"<<g;   return 0;
}

```

```

Output
Enter your name:
Saanvi
Enter your group:
2022
Saanvi
2022

=== Code Execution Successful ===

```

```

#include <iostream> using namespace std; int main()
{   char str[20];   int g;
    cout<<"Enter your name:\t";
    cin>>str;   cout<<"Enter your
group:\t";   cin>>g;
    cout<<str<<"\t"<<g;

```

```

Output
Enter your name:   Saanvi
Enter your group: 2022
Saanvi 2022

=== Code Execution Successful ===

```

```

#include <iostream> #include<iomanip> using
namespace std; int main() {   char str[20];   int g;
    cout<<"Enter your name:\b";
    cin>>str;   cout<<"Enter your
group:\b";   cin>>g;
    cout<<str<<setw(10)<<g;

```

```

Output
Enter your name: Saanvi
Enter your group: 2022
Saanvi 2022

=== Code Execution Successful ===

```

```

    return 0; }

```

```

    return 0;
}

#include <iostream> using namespace std;
int main() {
    char str[20]; int g;
    cout<<"Enter your name:"<<endl; cin>>str;
    cout<<"Enter your group:" <<endl; cin>>g;
    cout<<str<<endl<<g;

```

```

#include
<iostream>
namespace std;
int g;
    cout<<"Enter
cout<<"Enter

```

```

    return 0;
}

```

Output

```

Enter your name: Saanvi
Enter your group: 2022
Saanvi      2022

=== Code Execution Successful ===

```

Output

```

Enter your name:
saanvi
Enter your group:
2022
saanvi
2022

=== Code Execution Successful ===

```

```

#include<iomanip> using
int main() { char str[20];

    your name:\b"; cin>>str;
    your group:\b"; cin>>g;
    cout<<str<<setw(10)<<g;

```

Q6. WAP to utilize assignment += and -= type operators.

```

#include <iostream> using namespace std; int main() {

```

Output

```

The resultant number is:8

=== Code Execution Successful ===

```

```

    return 0;
}
int number=5;    number+=3;
cout<<"The resultant number is:",number;
#include <iostream>    using namespace std;
int main() {

    int number=5;    number-=3;    cout<<"The
resultant number is:"<< number;

return 0;
}

```

```

Output
The resultant number is:2
=== Code Execution Successful ===

```

Q7. WAP to swap two numbers using a bitwise operator.

```

#include <iostream>    using namespace std;    int main()    {        int a,b;

        cout<<"Enter a number: a = ";        cin>>a;        cout<<"Enter a number: b = " ;

cin>>b;        a=a^b;        b=a^b;        a=a^b;

```

```

Output
Enter a number: a = 5
Enter a number: b = 6
a is = 6
b is = 5

=== Code Execution Successful ===

```

```

        cout<<"a is = "<<a<<endl;
cout<<"b is = "<<b<<endl;

return 0;

}

```

Q8. Write a program to solve the following problem

A library charges a fine for every book return late. For first 5 days the fine is 50 paise, for 6-10 days fine is one rupee and above 10 days fine is 5 rupees. If you return the book after 30 days your membership will be cancelled .WAP to accept no. of days the member is late to return the book and display the fine or appropriate message.

```

#include <iostream> using
namespace std;

int main()
{   int days;

    cout << "Enter the number of days: ";   cin >>
days;

    if (days > 0 && days < 6)
    {       cout << "The fine is 50
paise";
    }
    else if (days >= 6 && days < 11)
    {
        cout << "The fine is 1 rupee";
    }
}

```

```

Output
Enter the number of days: 7
The fine is 1 rupee

=== Code Execution Successful ===

```



```

else if (days >= 11 && days < 31)
{
    cout << "The fine is 5 rupees";
} else if (days >= 31)
{
    cout << "YOUR MEMBERSHIP WILL BE CANCELLED";
}
else
{
    cout << "Undefined day";
}

return 0;
}

```

```

Output
Enter the number of days: 6000
YOUR MEMBERSHIP WILL BE CANCELLED

=== Code Execution Successful ===

```

Q9. WAP to display arithmetic operations using switch-case statement.

```

#include <iostream> using
namespace std;

int main()
{
    int a,
    b;    char
    op;

    cout << "Enter two numbers: ";
    cin >> a >> b;

```

```

cout << "Enter operator (+, -, *, /): ";  cin >> op;

switch (op)
{
    case '+':
        cout << "Addition = " << a + b;
        break;
        case '-'
':
        cout << "Subtraction = " << a - b;
        break;
        case
'*':
        cout << "Multiplication = " << a * b;
        break;
        case '/':      if (b != 0)
        cout << "Division = " << a / b;
        else
        cout << "Division by zero not allowed";
        break;

default:
        cout << "Invalid operator";
    }

    return 0;
}

```

Output

```

Enter two numbers: 7 8
Enter operator (+, -, *, /): +
Addition = 15

=== Code Execution Successful ===

```

Q10. WAP to check whether the given number is even or odd (By using if-else and conditional operator).

```
#include <iostream> using namespace std; int main() { int a;
```

```
cin>>a;
```

```
{
```

```
    cout<<"Number is even ";
```

```
}
```

```
else
```

```
    cout<<"Number is odd";
```

```
    return 0;
```

```
}
```

Output

Enter a number: 7

Number is odd

=== Code Execution Successful ===

Output

Enter a number: 6

Number is even

=== Code Execution Successful ===

```
cout<<"Enter a number: ";
```

```
if(a%2==0)
```